Record-Surface Visibility (RSV): Epistemic Limits of Administrative Export Artefacts

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Abstract

Administrative record systems are inherently relational, maintaining routing histories, parallel identifiers, version structures, assignment logs, and cross-channel interactions that shape the lifecycle of records. Yet external observers — requesters, oversight bodies, journalists, and parties to administrative disputes — rarely encounter these systems directly. Instead, they access administrative processes through exported documentary artefacts such as PDF bundles, cumulative email chains, and static disclosure packages.

This paper introduces Record-Surface Visibility (RSV) as an analytic vocabulary for describing the observable relational properties of such artefacts. Rather than adjudicating whether relational structures are lost within originating systems — a claim requiring backend access that external observers do not possess — RSV describes the conditions under which those structures are observable, inferable, or indeterminate at the exported artefact surface. Employing a reflexive black-box analysis of a 250-page administrative disclosure corpus, the study identifies a structural condition termed representational contraction: the reduction of relational multiplicity into a stabilised documentary form that forecloses interpretive reconstruction. Significantly, even communications originally authored by the requester cannot be reliably reconstructed in their procedural context from the exported artefact.

RSV is proposed as an analytic vocabulary for articulating the epistemic limitations imposed by contemporary export infrastructures, with implications for archival theory, transparency scholarship, and the design of disclosure systems.

Keywords: record-surface visibility; administrative transparency; records management; epistemic observability; documentary export artefacts; representational contraction; documentary infrastructure

1. Introduction

Digital administrative systems are not neutral repositories of stored information. They are constitutively relational infrastructures that sustain routing histories, maintain parallel identifiers, record version structures, log assignment changes, and preserve cross-channel interactions.

These relational properties are not incidental to the records they hold. They constitute the primary architecture through which institutional memory is sustained and administrative decisions can, in principle, be reconstructed, scrutinised, and contested.

The accountability value of administrative records resides not merely in their informational content, but in the legibility of the procedural relations that produced them.

External observers — whether requesters pursuing access-to-information rights, oversight bodies exercising audit functions, journalists investigating institutional conduct, or parties to administrative disputes — rarely encounter this relational architecture directly. Their access to administrative processes is mediated through documentary export artefacts: PDF bundles produced by case management systems, email chains rendered through disclosure processes, compressed ZIP archives assembled for formal release, and static disclosure packages constructed in response to information requests. These artefacts constitute the de facto evidentiary surface through which administrative action is interpreted by those outside the producing institution.

This creates a fundamental epistemic asymmetry. The institution that produces the export operates with full knowledge of the relational architecture underlying it. External observers can access only what is visible at the documentary surface. Whether formally complete disclosures are also epistemically adequate therefore depends on whether the relational context necessary for interpretation is observable at the exported artefact surface.

Existing scholarship in archival science has examined metadata reduction, contextual integrity, and record transformation across systems (Duranti, 1998; Yeo, 2007, 2018). This body of work has predominantly focused on system-internal processes — the mechanisms by which relational structure is altered, degraded, or preserved within institutional systems. Complementary work in diplomatics has examined authenticity and evidential form in digital environments (Duranti, 2009), and infrastructure studies have illuminated how technological form shapes interpretability (Bowker and Star, 1999; Dourish, 2017). The transparency

literature, meanwhile, has examined formal compliance with disclosure regimes while leaving largely unaddressed the structural opacity that formally complete disclosures may nonetheless impose (Roberts, 2006; Worthy, 2017).

A gap persists, however, at the intersection of these literatures: the downstream epistemic condition of external observers who must interpret exported outputs without access to originating infrastructures has not been systematically theorised. A disclosure may be formally complete while remaining structurally opaque, insofar as the relational context necessary for interpretation is not observable at the documentary surface. This paper addresses that gap.

From an archival perspective, this study is concerned with how records function as evidence once they are encountered at the documentary surface. Archival theory has long emphasised the importance of provenance, archival bonds, and contextual integrity in sustaining the evidential value of records. When records are accessed through exported documentary artefacts rather than system-native environments, however, the visibility of these relational properties becomes uncertain. RSV is proposed as an analytic vocabulary for describing evidential visibility at precisely this juncture, focusing on the observable relational properties of documentary export artefacts and the conditions under which such properties are observable, inferable, or indeterminate at the documentary surface.

The vocabulary is developed through a reflexive black-box analysis of a 250-page administrative disclosure corpus. The analysis identifies a structural condition termed representational contraction, in which relational multiplicity is reduced to a stabilised documentary form from which procedural context cannot be reliably reconstructed. The contribution of this study lies not in empirical generalisation but in conceptual articulation: providing a structured analytic vocabulary for identifying and describing a structurally identifiable interpretive condition that may arise wherever access to administrative records is mediated through stabilised documentary outputs. The tripartite distinction among observable, inferable, and indeterminate structures underpins the vocabulary and is applied across each of its five dimensions. The paper proposes RSV as an analytic vocabulary with implications for archival theory, transparency scholarship, and the institutional design of export systems.

2. Theoretical Context

2.1 Records, Relational Context, and Evidential Form

Archival theory has consistently held that the evidential value of records depends not only on their informational content but on their contextual and relational properties. Records are understood as structured entities embedded in procedural contexts that confer upon them their status as evidence of administrative action (Yeo, 2007). This view has its intellectual roots in classical diplomatics, where the authenticity of a document was assessed through analysis of its formal elements, its relational position within a documentary series, and its conformity to the procedural forms of its producing institution (Duranti, 1998). In digital environments, this tradition has been extended through the InterPARES project's systematic analysis of the conditions under which digital records can be held to maintain their authenticity, completeness, and evidential integrity across system migrations and long-term preservation contexts (Duranti, 2009).

Yeo's conceptual analysis of records distinguishes between records as evidence, as information, and as persistent representations. For Yeo, the evidential function of records is inseparable from their contextual embeddedness: a record's capacity to serve as evidence of past action depends on its ability to anchor that action within a determinable procedural context (Yeo, 2007, 2018). This perspective foregrounds what may be called the relational dimension of records: the network of procedural dependencies, cross-references, assignments, and interactions that constitute the context in which a record was produced and through which its meaning can be assessed.

Digital records management has extended this framework to address the specific challenges posed by complex, relational data systems. Contemporary administrative systems — case management platforms, enterprise content management systems, workflow automation tools — maintain relational architectures of considerable complexity: routing histories that record how a document traversed an organisation; version histories that preserve the evolution of a record over time; assignment logs that document which actors acted upon a record and when; threading structures that preserve the relationships between communications across channels. These features collectively constitute the relational context from which a record's administrative significance can be reconstructed.

The Records Continuum model, developed by Upward (1996, 2000) and elaborated through subsequent archival scholarship (McKemmish, 2001), offers a further theoretical resource for situating this study. Where traditional lifecycle models treat records as passing through

discrete sequential stages, the continuum model conceptualises records as existing simultaneously across multiple dimensions of space, time, and purpose. Within this framework, relational context is not a static attribute attached to a record at creation but a dynamic property constituted and reconstituted through ongoing archival processes. RSV does not operate within the continuum model's scope — it is concerned with a specific moment of documentary fixation rather than with ongoing relational constitution — but the continuum framework illuminates what is at stake in that moment. Export is the point at which the dynamic relational multiplicity of the continuum is arrested into a static documentary form. RSV describes what of that relational multiplicity remains visible at the arrested surface, and what does not.

2.2 Export as Infrastructural Transformation

Despite the richness of relational architectures within contemporary administrative systems, external access to those systems is almost universally mediated through exported documentary artefacts. This mediation has received less systematic theoretical attention than it deserves. Export is typically treated in the literature as a technical process, the conversion of data from one format to another, rather than as an analytically significant infrastructural transformation with epistemically consequential effects.

From an infrastructure studies perspective, however, this characterisation is insufficient. As Bowker and Star observe, infrastructure becomes visible primarily at moments of breakdown (1999). Under normal operating conditions, it recedes from view, shaping the conditions of practice without itself being an object of attention. The export process, similarly, functions as an invisible infrastructure when it produces artefacts that are treated as transparent windows onto the administrative systems they purport to represent. But the export format is not a neutral medium of transmission; it is an active shaper of what can be known from the artefact it produces.

Dourish's account of the materiality of information provides a useful theoretical anchor here. For Dourish, information is not pre-given content that merely awaits a carrier; rather, the material form of information, the infrastructure through which it is encoded, stored, and circulated, is constitutive of the interpretive possibilities it affords (Dourish, 2017). Applied to export artefacts, this perspective implies that the format of a documentary export is not merely incidental to its content; it determines what features of the originating system can be

represented, which relations can be preserved, and what interpretive operations the resulting artefact can support.

Contemporary export formats — particularly PDF and PDF/A — are designed to prioritise visual stability, long-term fixity, and format-independent legibility (ISO, 2020). These design priorities are well-suited to archival preservation goals, but they come at a cost to relational functionality. A PDF export may faithfully reproduce the textual content of a record while rendering invisible the threading relationships, assignment histories, and routing structures that would enable an external observer to reconstruct its procedural context. This tension between preservation as content fidelity and preservation of relational legibility is not merely a technical trade-off; it is an epistemic one with direct consequences for the accountability functions that disclosure is intended to serve.

2.3 Transparency, Disclosure, and Structural Opacity

The transparency literature has extensively examined the operation of access-to-information frameworks and the conditions under which formal disclosure translates into substantive accountability (Roberts, 2006; Worthy, 2017). A consistent finding in this literature is the gap between formal and substantive transparency: a disclosure may be formally compliant, responsive to the terms of the request and complete in the records it releases, while remaining substantively opaque insofar as the released records do not enable external observers to understand the administrative processes they document.

Roberts' analysis of governmental secrecy demonstrates how institutional resistance to transparency can manifest not only through explicit withholding but through the management of documentary outputs: the selection, packaging, and format of released materials can function as instruments of opacity even when the materials themselves are formally released (Roberts, 2006). Worthy's account of freedom of information politics similarly identifies the management of disclosure formats as a locus of institutional agency in shaping the interpretive possibilities available to requesters (Worthy, 2017).

This literature, however, has not developed systematic conceptual tools for analysing the specific ways in which export formats shape the relational observability of documentary artefacts. The contribution of RSV lies precisely here: in providing a descriptive vocabulary for articulating the mechanisms by which export infrastructure determines what is

epistemically available to external observers at the documentary surface, independently of questions about institutional intent or formal compliance.

It is worth distinguishing RSV from adjacent concepts with which it might be conflated. Contextual integrity, as developed by Nissenbaum (2004) and applied to information flows, addresses the normative appropriateness of information transmission across social contexts; RSV is not a normative framework but a descriptive one, and its concern is observability rather than appropriateness. The records management literature's treatment of appraisal and arrangement addresses decisions about which records to retain and how to organise them within archival holdings (Gilliland, 2014; Shepherd and Yeo, 2003); RSV addresses neither retention decisions nor archival arrangement, but the epistemic properties of exported outputs as encountered by external observers. Metadata standards scholarship addresses the design of metadata schemas to support interoperability and long-term usability (ISO, 2020); RSV addresses what is actually observable at the documentary surface of a given artefact, not what metadata schemas prescribe. RSV therefore occupies a distinct conceptual space: it is a vocabulary for surface-level epistemic description, not a normative, managerial, or design framework.

3. Methodology and Positionality

3.1 The Black-Box Perspective

The analysis adopts a strict black-box perspective: only observable surface characteristics of the documentary artefact are examined. These include layout structure and visual organisation; identifier presentation and labelling conventions; visible metadata elements; repetition patterns and nested structures; navigational affordances and reference mechanisms; and formatting conventions that may carry procedural significance. The internal architecture of the originating administrative system — its data model, relational schema, workflow logic, and system-level metadata — is neither assumed nor verified.

This methodological choice is not merely a practical constraint but a principled one that directly enacts the epistemic position RSV seeks to describe. External observers in disclosure contexts do not have access to the originating system; they encounter only the artefact produced by the export process. A methodology that drew on backend system knowledge to interpret documentary surfaces would misrepresent the epistemic situation of the actors for whom RSV is designed as an analytic tool. The black-box perspective is therefore not an

approximation of a more complete method but the methodologically correct approach for analysing observability from the external observer's standpoint. RSV must be grounded exclusively in what is observable from that position.

Accordingly, the absence of observable relational features is interpreted throughout this analysis as a condition of non-observability rather than as evidence of system-level absence or failure. RSV describes epistemic limits, not infrastructural deficits. The distinction matters: a record management system may maintain full relational integrity internally while producing exports from which that integrity is not recoverable. RSV identifies and characterises the latter condition without purporting to adjudicate the former.

3.2 The Corpus

The empirical material comprises a 250-page documentary disclosure corpus produced in response to an administrative information request. The corpus consists primarily of email communications, supplemented by associated administrative records and procedural documents. It was assembled and released through a formal administrative disclosure process.

The corpus is treated as a bounded and reflexive case. It is not deployed as a basis for generalisable claims about the properties of administrative export artefacts in general, but rather as a site for the detailed development and illustration of the RSV analytic vocabulary. Its value for this purpose derives precisely from its specificity and from the internal reference point it provides, described below.

This methodological choice is grounded in the logic of theoretical case selection rather than representational sampling. As Glaser and Strauss (1967) established in the context of grounded theory, and as Yin (2014) has elaborated for case study research, a single case can serve legitimate theoretical purposes when it is selected not to represent a population but to illuminate, develop, or challenge conceptual categories. The present case functions in this mode: it is selected because it affords unusually precise conditions for examining the epistemic limits of record-surface observability, not because it is typical of administrative export artefacts in general. The contribution of the analysis is the vocabulary it generates and the structural condition it makes analytically visible, not a statistical claim about prevalence. The single-case design is therefore not a limitation on ambition but a feature of the methodology's purpose.

3.3 Reflexive Positionality and the Self-Authored Records Problem

The analysed corpus presents an analytically significant circumstance that warrants explicit attention: the corpus includes communications originally authored by the requester. This creates an internal reference point for assessing the limits of record-surface observability that would not be available in a purely third-party disclosure corpus.

The expectation one might hold is that communications authored by the requester would be straightforwardly recoverable from the disclosure corpus: the requester possesses independent knowledge of their authorship, their content, and their procedural context. The RSV analysis of this corpus reveals, however, that even self-authored records cannot be reliably reconstructed in their procedural context from the exported artefact. The identity of messages across different appearances in the corpus is not determinable; the sequential relationships between communications cannot be established; the procedural occasions of authorship — who a message was addressed to, in what capacity, at what procedural stage — are not observable at the documentary surface.

This finding provides a methodologically rigorous lower bound on record-surface observability. If procedural context is non-recoverable for records whose content and authorship are independently known to the observer, the condition of non-observability cannot be attributed to observer unfamiliarity with the subject matter. It is a structural property of the export artefact itself.

A potential concern with the reflexive design merits direct address: whether the researcher's dual position as requester and analyst introduces a risk of motivated or partial analysis. This concern is methodologically resolved by the black-box constraint. The analytical object is restricted to surface-visible features of the documentary artefact — layout, identifier presentation, metadata elements, navigational affordances — none of which depend on the researcher's prior knowledge to identify or describe. Crucially, the researcher's independent knowledge of the self-authored records does not enter the analysis as evidence; it functions only as a baseline against which the limits of surface observability are measured. The finding is not 'the researcher knows more than the surface shows' — which would be trivially true of any observer with independent knowledge — but 'the surface does not provide sufficient features to recover procedural context even for records whose content is independently known.' That finding is derivable from surface observation alone, and the researcher's prior knowledge merely makes its significance analytically legible.

This study analyses a corpus of administrative records obtained through a formal public disclosure process and available to the requester as a matter of administrative right. The materials are documentary in nature and do not contain data collected from human subjects in a research context. No personal data beyond that ordinarily present in formal administrative correspondence is involved, and no third parties are identified or discussed in the analysis. The study accordingly does not raise ethics concerns requiring institutional review board oversight, consistent with standard practice for archival and documentary research of this kind.

The analysis is strictly confined to observable documentary features. It does not evaluate institutional intent, assess system design choices, or draw inferences about the motivations of producing actors. Questions of intent and institutional conduct are beyond both the scope of RSV and the methodological limits of a black-box analysis.

4. The RSV Analytic Vocabulary

RSV provides a structured descriptive vocabulary for articulating the conditions under which relational properties of administrative records may or may not be determinable at the documentary surface. It is organised around five dimensions, each targeting a distinct aspect of relational observability. The dimensions are descriptive rather than evaluative: they characterise what is observable, not what should be observable, and they make no presupposition about what a well-formed export artefact ought to look like. The vocabulary is offered as an analytic tool for external observers seeking to characterise the epistemic limits imposed by the artefacts they are analysing. The five dimensions proposed here are not intended as an exhaustive or closed taxonomy; they are offered as a provisional framework that may be refined, extended, or challenged through comparative application across different disclosure contexts and administrative settings.

4.1 Identifier Continuity

Identifier Continuity addresses the most fundamental question of record individuation: whether the same record, appearing at multiple locations within an export corpus, can be recognised as the same record. This dimension is analytically prior to the others — without stable record identity, questions of metadata, pathway, attachment, and integration cannot be anchored to determinate documentary instances. Identifier Continuity refers to the capacity to track the identity of a record across multiple appearances within the export corpus.

Administrative systems typically assign persistent identifiers to records — case numbers, ticket identifiers, message identifiers, version numbers — that enable records to be tracked across different views, states, and locations within the system. At the documentary surface, the question of Identifier Continuity is whether such identifiers are present, consistently rendered, and sufficient to establish cross-reference relationships between different documentary instances.

In the analysed corpus, Identifier Continuity was consistently low. Email messages appearing at multiple points in the corpus lacked stable surface-visible identifiers that would permit their reliable identification across instances. In nested quotation structures, earlier messages were reproduced without identifiers that would allow them to be linked to their standalone appearances elsewhere in the corpus. The result is that, at the documentary surface, individual records cannot be reliably individuated as the same record across contexts: it is not determinable whether two instances represent the same message or two distinct messages with similar content.

4.2 Metadata Exposure

Where Identifier Continuity concerns the capacity to track records across the corpus, Metadata Exposure concerns the richness of the procedural information attached to each record at its individual appearances. Metadata Exposure refers to the extent to which procedurally significant metadata elements are rendered visible at the documentary surface. In native administrative systems, records are typically associated with rich metadata: timestamps, routing information, sender and recipient identifiers, read receipts, system-generated status fields, procedural labels, and classification tags. These metadata elements are often as important as content for interpreting a record's procedural significance — when it was created, by whom, in what capacity, at what stage of a workflow, and how it was received.

At the exported artefact surface, metadata exposure is typically partial and variable. Some metadata elements — sender identity, nominal timestamp, subject line — are ordinarily rendered visible as part of the standard email header display format. Others — routing history, system identifiers, version information, read receipt status, procedural classification — are frequently absent from the surface rendering. The RSV dimension of Metadata Exposure tracks the specific pattern of present and absent metadata elements, with attention to the procedural significance of what is and is not visible.

In the analysed corpus, Metadata Exposure was found to be structurally selective. Header-level metadata (sender, nominal timestamp, subject) was consistently present; routing, procedural, and system-level metadata was not surface-visible. This selectivity was consequential: without routing metadata, for example, it was not possible to determine from the documentary surface whether a communication had been forwarded, assigned, or processed through particular workflow stages before reaching its final destination in the corpus.

4.3 Pathway Representation

Where Metadata Exposure addresses the richness of information attached to a record at a given documentary instance, Pathway Representation addresses whether the sequence of procedural events that produced that instance can be reconstructed across the corpus as a whole. Pathway Representation refers to the observability of the procedural pathways through which a record has travelled: the routing, assignment, escalation, and handling history that constitutes the record's administrative biography. In relational administrative systems, pathway information is maintained through a combination of routing metadata, assignment logs, workflow state records, and threading structures. This information is often critical for accountability purposes: it establishes which actors handled a record, in what capacity, and at what stages of an administrative process.

At the exported artefact surface, pathway information is rarely directly represented. The artefact typically records the end-state of a communication chain — its textual content at a given moment — without making visible the procedural journey that produced that end-state. An email appearing in a disclosure corpus may have been forwarded multiple times, assigned across multiple case handlers, or processed through a series of administrative workflow states before its export; none of this history may be recoverable from the documentary surface alone.

The RSV dimension of Pathway Representation therefore examines whether the sequential and relational structure of administrative pathways can be reconstructed from observable surface features. In the analysed corpus, pathway reconstruction was not possible from the documentary surface: the routing, assignment, and handling history of communications could not be determined. The corpus records outcomes without representing the processes that produced them.

4.4 Attachment Lineage Visibility

Where Pathway Representation concerns the procedural history of communications as records, Attachment Lineage Visibility concerns the integrity of the documentary units formed by those communications and the files associated with them. Attachment Lineage Visibility refers to the observability of the relationships between a communication and any documents, files, or attachments associated with it in the originating system. In administrative systems, attachments are relationally integrated with their parent communications: they are referenced, transmitted, versioned, and processed as part of a documentary unit whose integrity depends on the traceability of the attachment relationship. At the documentary surface of an export artefact, the question is whether this relational integrity is observable.

Three aspects of Attachment Lineage Visibility can be distinguished. First, presence observability: it may or may not be determinable from the documentary surface whether an attachment existed in association with a given communication. Second, identity observability: where an attachment is surface-visible, it may or may not be possible to determine which specific file version was associated with a communication, particularly where multiple versions of a document exist. Third, threading observability: where an attachment is reproduced across multiple instances within the corpus, it may not be determinable whether these instances represent the same attachment or distinct versions of a similarly named document.

In the analysed corpus, Attachment Lineage Visibility was low across all three dimensions. The presence of attachments in the original communications could not be reliably inferred from the export format; where documents appeared in the corpus, their relationship to specific communications was not traceable; and where similar documents appeared at multiple points, their identity or distinctness could not be established from surface-visible features.

4.5 Format Integration

The preceding four dimensions address properties of individual records or record-to-record relationships within the corpus. Format Integration addresses a higher-order property: the coherence of the export artefact as a navigable whole. Format Integration refers to the coherence of the export artefact as a navigable, cross-referenced documentary unit. Contemporary administrative disclosure corpora frequently comprise records drawn from

multiple originating systems — email platforms, document management systems, case management tools, calendar applications — that are assembled into a single export package. At the documentary surface, the question is whether this assemblage maintains sufficient internal cross-referencing to permit navigation across its constituent parts.

Format Integration encompasses internal navigability (the capacity to move between related documents within the corpus using surface-visible references), cross-reference visibility (the observability of references between records in the corpus), and format consistency (the extent to which records from different originating systems are rendered in a format that permits comparative reading). Low Format Integration produces a corpus that can be read linearly but cannot be navigated relationally: cross-references cannot be followed, related records cannot be grouped, and the structural relationships between records within the corpus are not determinable from its surface.

In the analysed corpus, Format Integration was minimal. The corpus was navigable only linearly; internal cross-references were not surface-visible; and records from different originating contexts were rendered in a format that did not expose their relational connections.

5. Case Illustration: Patterns of Representational Contraction

Application of the RSV vocabulary to the 250-page disclosure corpus yielded a consistent pattern of low relational observability across all five dimensions. This section provides illustrative characterisation of the key patterns identified, employing the RSV vocabulary to describe rather than evaluate the observed documentary conditions.

5.1 Nested Quotation and Identifier Absence

The most pervasive structural feature of the corpus was the rendering of email communications in cumulative nested quotation chains. In standard email client rendering, a reply chain produces a document in which successive messages are reproduced as quoted text within each subsequent reply. When such chains are exported as static documents, the result is a highly redundant corpus in which identical email content may appear numerous times, each time as a fragment within a larger message structure.

In the analysed corpus, identical textual content appeared across a large number of pages as quoted text within successive reply messages. In the absence of stable surface-visible

message identifiers, it was not possible to determine: whether a given instance of text represented the original transmission of that text or a subsequent quotation of it; how many distinct messages were represented by similar textual fragments; or what the sequential relationships between messages were within any given chain.

The RSV dimension of Identifier Continuity captures this pattern: the export format does not expose message-level identifiers that would permit disambiguation of instances. The result is not information loss in the conventional sense — the textual content of communications is abundantly present — but a form of epistemic indeterminacy in which the procedural context of that content is non-reconstructable.

Not all surface conditions in the corpus were conditions of strict indeterminacy. In isolated instances, relational structures were inferable from indirect surface cues even where they were not directly observable. For example, the presence of a forwarding header notation within an email rendered in the corpus permitted a reasonable inference that the message had passed through at least one additional recipient before reaching its documented addressee, even though no routing metadata was surface-visible. Such cases illustrate the inferable category: the relational structure cannot be directly evidenced from the surface, but its presence is reasonably deducible from indirect cues. Significantly, inferability of this kind was the exception rather than the condition; the predominant documentary condition was indeterminacy, in which no such inferential basis was available.

5.2 The Self-Authored Records Finding

The analytically most significant finding arose from the reflexive dimension of the case: the inability to reconstruct self-authored records in their procedural context. Communications originally produced by the requester appeared within the corpus in forms that did not permit reliable identification as self-authored, reconstruction of the procedural occasions of authorship, or determination of sequential and relational position within communication threads.

This finding is analytically significant because it precludes explanations of non-observability that depend on the observer's unfamiliarity with the subject matter. A researcher encountering an unfamiliar institutional domain might reasonably be expected to struggle with interpreting procedural context; the self-authored records finding demonstrates that procedural context is non-recoverable from the documentary surface even for records whose content the observer

independently knows. The non-observability documented by RSV is a property of the export artefact, not of the observer.

5.3 Representational Contraction

The aggregate effect of the patterns documented across the five RSV dimensions is captured in the concept of representational contraction. Representational contraction denotes the reduction of relational multiplicity — the rich, multi-dimensional relational context maintained within an administrative system — into a stabilised documentary form that forecloses the interpretive reconstruction of that context by external observers.

Representational contraction does not require that relational structures be destroyed within the originating system. It is consistent with the possibility that the system maintains full relational integrity internally. What representational contraction describes is the condition of the documentary surface: the extent to which the relational architecture of the originating system is rendered non-observable by the export format. A system may contract its relational architecture into a surface-stable documentary representation without that architecture being absent from its own records.

RSV does not resolve the condition of representational contraction. Its contribution is to make it analytically visible and articulable: to provide a vocabulary through which external observers can describe, with precision, what they cannot know and why.

6. Discussion

6.1 RSV as Epistemic Vocabulary

The theoretical contribution of RSV lies in its reorientation of the analytical problem. Archival theory has long foregrounded provenance, archival bonds, and contextual integrity as the conditions under which records sustain their evidential value. These concepts address the relational architecture of records as it is constituted and maintained within institutional systems. What they do not address is the condition of external observers who encounter records only at the documentary surface — who cannot inspect provenance chains, verify archival bonds, or assess contextual integrity from within the originating system, but must instead determine, from surface-visible features alone, whether and to what extent these relational structures are present, inferable, or indeterminate.

RSV occupies this gap. By foregrounding the documentary surface as a distinct site of analysis and adopting the epistemic position of external observers as its analytical standpoint, RSV does not compete with or replace existing archival frameworks. It extends them to a domain — the exported artefact as encountered by those outside the producing institution — that those frameworks have not systematically addressed. The shift is from what has happened to the data to what can be known from the artefact; from system-internal integrity to surface-level observability.

The concept of representational contraction warrants a note on its theoretical status. It is introduced in Section 5 as an observation arising from the case analysis, but it is offered here as a theoretical concept with broader applicability: a name for the structural condition in which the relational multiplicity of an administrative system is arrested into a documentary form that does not render that multiplicity observable to external observers. Representational contraction is not an empirical generalisation from the present case — the present case instantiates it, but does not establish its prevalence. It is a conceptual category proposed to make a recurring interpretive condition analytically articulable. Its value lies not in frequency claims but in precision: it distinguishes the condition of surface indeterminacy from information loss, from institutional withholding, and from observer incompetence, identifying it instead as a structural property of the relationship between relational systems and their documentary exports.

This shift has practical as well as theoretical significance. For requesters and oversight bodies, RSV provides a structured vocabulary for articulating the limits of what a disclosure enables them to know — a vocabulary that does not require insider knowledge to apply, and that makes analytically precise what might otherwise remain an inchoate sense that a formally complete disclosure is nonetheless difficult to interpret. For archival theorists, RSV identifies a category of epistemic condition — surface indeterminacy — that is not reducible to information loss and requires its own conceptual treatment. The condition is one in which informational content remains accessible at the documentary surface while relational context is not: the records are readable but their procedural context is not recoverable. For institutional designers, the five dimensions of RSV identify specific axes along which export formats shape the epistemic conditions of accountability. In this respect, RSV extends archival discussions of records as evidence by identifying a distinct epistemic condition in which evidential interpretation is constrained not by the absence of records, but by the limits of their surface-level observability.

6.2 Format, Infrastructure, and Accountability

The findings of this analysis have implications beyond archival theory. They speak to a broader problem in the institutional design of disclosure systems: the assumption that formally complete disclosure produces substantively adequate accountability. RSV demonstrates that this assumption may be structurally undermined by export infrastructures that reduce relational observability at the documentary surface.

This is not, in the first instance, a claim about institutional intent. Export formats are often adopted for reasons that have nothing to do with accountability management: PDF/A is the standard format for long-term preservation because of its fixity properties, not because of any intention to reduce relational observability. The epistemic effects of format choices may be unintended consequences of design decisions taken with other priorities in mind. Nevertheless, those effects are real, and they have consequences for the accountability functions that disclosure is intended to serve.

The implication is that accountability-oriented analysis of disclosure systems should attend not only to the completeness of the records released but to the relational observability of the formats in which they are released. A fully responsive disclosure that produces an artefact from which procedural context is non-recoverable may satisfy formal requirements while failing substantive ones.

6.3 Limitations and Scope

RSV is offered as an exploratory vocabulary rather than a fully elaborated theoretical framework. Several limitations bear acknowledgement. The empirical basis of this study is a single bounded corpus, and the analytic vocabulary has been developed through close engagement with that corpus rather than through systematic comparison across multiple disclosure contexts. As noted in Section 3, this design is theoretically motivated rather than merely practical; the case was selected for its analytic properties rather than its representativeness. Nevertheless, the vocabulary's applicability across different administrative contexts, disclosure formats, and jurisdictional settings remains a question for future comparative research.

While developed from a single disclosure corpus, the RSV framework is expected to be applicable to any context in which access to administrative records is mediated through

stabilised documentary exports. Such research would be well positioned to test, refine, and potentially extend the five dimensions proposed here.

The black-box perspective, as argued in Section 3, is the methodologically appropriate standpoint for analysing the epistemic condition of external observers. It does not permit claims about what is present or absent within originating systems, and RSV makes none. Future research with access to both the originating system and its exported outputs could productively examine the relationship between system-level relational integrity and surface-level observability — a question that lies beyond RSV's scope but would enrich understanding of the export process as an infrastructural transformation.

Finally, RSV as developed here is a descriptive vocabulary rather than a prescriptive standard. It does not specify what adequate relational observability would look like, nor does it propose criteria for evaluating whether a given export artefact meets accountability requirements. These normative questions are important but beyond the scope of the present study.

7. Conclusion

This paper has introduced Record-Surface Visibility (RSV) as an analytic vocabulary for describing the observable relational properties of administrative documentary export artefacts. Grounded in a reflexive black-box analysis of a 250-page disclosure corpus, and developed in dialogue with archival theory, infrastructure studies, and transparency scholarship, RSV offers a structured vocabulary for articulating the epistemic limits imposed on external observers by contemporary export infrastructures.

The five dimensions of RSV — Identifier Continuity, Metadata Exposure, Pathway Representation, Attachment Lineage Visibility, and Format Integration — provide a framework for characterising the specific ways in which relational observability is constrained at the documentary surface. Each dimension tracks a distinct aspect of the observable/inferable/indeterminate spectrum, and together they capture the overarching condition of representational contraction: the reduction of relational multiplicity into a stabilised documentary form from which procedural context cannot be reliably reconstructed. The finding that this condition applies even to self-authored records provides a methodologically robust lower bound on surface observability, establishing that the limits documented by RSV are properties of the artefact rather than of the observer.

The contribution of this study lies not in empirical generalisation but in conceptual articulation. By foregrounding the documentary surface as a distinct site of analysis — analytically separable from both the system-internal records management questions addressed by archival theory and the formal compliance questions addressed by transparency scholarship — RSV introduces a new analytical unit and a new observational standpoint. This is not an extension of existing archival discussions so much as an identification of a domain those discussions have not yet inhabited: the epistemic condition of external observers who encounter administrative records only at the stabilised documentary surface, and who must determine from that surface alone what can and cannot be known about the processes the records purport to document.

RSV does not resolve the epistemic conditions it describes. It makes them analytically visible and articulable. In doing so, it opens conceptual space for further inquiry into the infrastructural dimensions of accountability and transparency, the epistemic implications of archival format choices, and the design of export systems that take relational observability seriously as a value alongside fixity and legibility. The vocabulary is offered as a contribution to that inquiry and as a documentation of structural patterns that, in the absence of an adequate descriptive apparatus, may otherwise remain unnamed. In this sense, RSV contributes to archival theory by shifting attention from the preservation of records to the conditions under which they remain evidentially interpretable at the documentary surface.

RSV is offered as a provisional analytic vocabulary intended to support further comparative application and conceptual refinement across different administrative and disclosure contexts.

Appendix A — Archival Records

Supporting archival records are preserved in an external repository (Zenodo) and are accessible via persistent identifiers. These materials are maintained as DOI-linked research archives and resolve to the most recent available version of each item.

They are provided to support documentation transparency and to strengthen the evidential grounding of the empirical context discussed in this study.

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